

Getting Started with Script Extender

Getting started with Script Extender

This tutorial covers the absolute basics for modders to start writing mods with [Norbytes Script Extender](#) [\[E\]](#).

No prior knowledge about programming is required; however, please familiarize yourself with how to install mods and Script Extender:

- [How to install .pak files](#)
- [Installing Script Extender and activating the console](#)

After reading this guide, feel free to look at the example [Changing an entity's name](#) or follow the tutorial: [Creating your First SE Mod \(TBA\)](#).



If you are having trouble following this guide, feel free to join the Baldur's Gate 3 Modding Community discord linked [here](#) [\[E\]](#). The Baldur's Gate 3 Modding Community is a community modding run discord. Keep in mind that everyone that may help you is a volunteer. Please read and follow the server rules.

You can find further information linked under [09. Useful Resources](#) on the bottom of this page.

1. Activate the Script Extender Console

After following the guide [How to install Script Extender](#) and activating the console, ensure that the console works properly.

Launch your game and when the game exe opens, a second window will also open. When you launch the game now, you should see a second window opening. This is the Script Extender console.

```

BG3 Script Extender Debug Console
-----
BG3 Script Extender Debug Console
-----
BG3Ext v16 built on Apr 30 2024 22:11:17
Game version v4.50.22.896 OK
ScriptExtender::Initialize: Starting
Library startup took 930 ms
We're already past App load, triggering CoreLibInit
LibraryManager::PostStartupFindLibraries() took 108 ms
u Initializing client with target context Load
Modded achievements enabled.
bg3se::ExtensionStateBase::LuaResetInternal(): LUA VM reset.
Initializing client with target context Game
bg3se::ExtensionStateBase::LuaResetInternal(): LUA VM reset.
ecl::ScriptExtender::OnGameStateChanged(): Loaded module

```

If you see this window, it means you have successfully installed SE with console.

2. Familiarizing yourself with the Script Extender Console

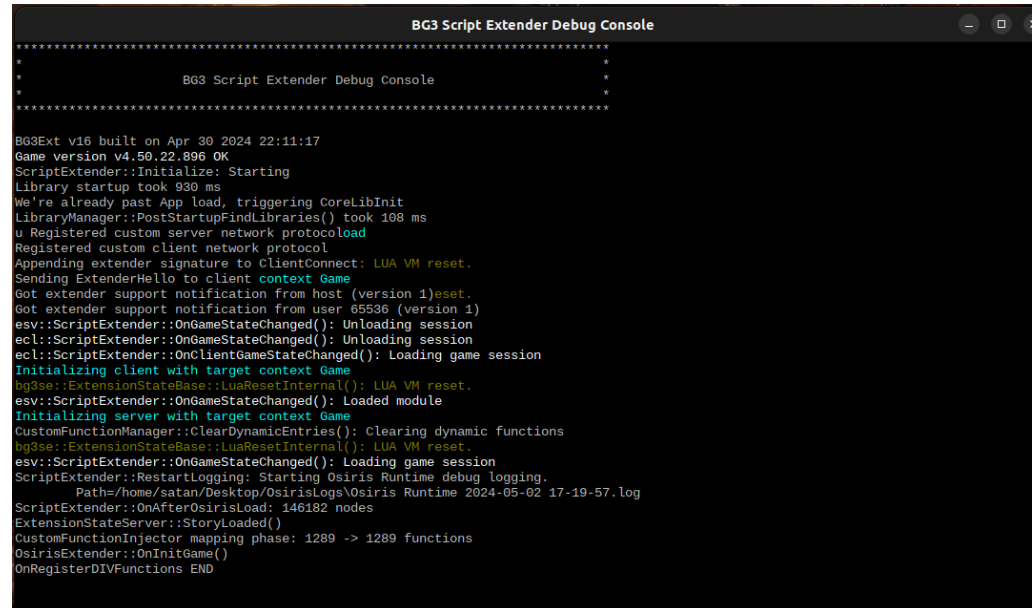
To familiarize yourself with the usage of the SE console you can execute a few simple commands.

These can all be found in the [09.Useful Resources](#) at the bottom of this tutorial as [Osi functions](#) [↗](#) .

2.1. Load a save

You can either start a new game or load an existing save.

When the save is loaded you should see some output in your console, telling you that the session has loaded and other debug statements depending on what kind of mods you have installed:



```

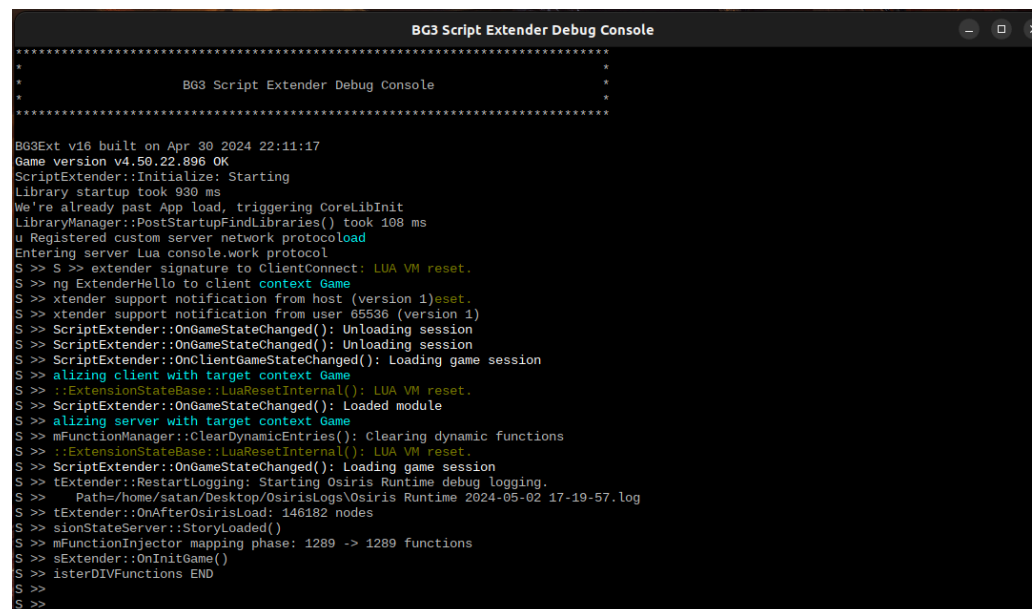
BG3 Script Extender Debug Console
*****
*                                     *
*          BG3 Script Extender Debug Console          *
*                                     *
*****

BG3Ext v16 built on Apr 30 2024 22:11:17
Game version v4.50.22.896 OK
ScriptExtender::Initialize: Starting
Library startup took 980 ms
We're already past App load, triggering CoreLibInit
LibraryManager::PostStartupFindLibraries() took 108 ms
u Registered custom server network protocolload
Registered custom client network protocol
Appending extender signature to ClientConnect: LUA VM reset.
Sending ExtenderHello to client context Game
Got extender support notification from host (version 1)reset.
Got extender support notification from user 65536 (version 1)
esv::ScriptExtender::OnGameStateChanged(): Unloading session
ecl::ScriptExtender::OnGameStateChanged(): Unloading session
ecl::ScriptExtender::OnClientGameStateChanged(): Loading game session
Initializing client with target context Game
hg3se::ExtensionStateBase::LuaResetInternal(): LUA VM reset.
esv::ScriptExtender::OnGameStateChanged(): Loaded module
Initializing server with target context Game
CustomFunctionManager::ClearDynamicEntries(): Clearing dynamic functions
hg3se::ExtensionStateBase::LuaResetInternal(): LUA VM reset.
esv::ScriptExtender::OnGameStateChanged(): Loading game session
ScriptExtender::RestartLogging: Starting Osiris Runtime debug logging.
Path=/home/satan/Desktop/OsirisLogs/Osiris Runtime 2024-05-02 17-19-57.log
ScriptExtender::OnAfterOsirisLoad: 146182 nodes
ExtensionStateServer::StoryLoaded()
CustomFunctionInjector mapping phase: 1289 -> 1289 functions
OsirisExtender::OnInitGame()
OnRegisterDIVFunctions END
  
```

2.2. Activate the input

To be able to input commands into the console press enter once.

If you have too much text on your console and you cannot see your input line, you can repeat this a few times until you can see the S >>



```

BG3 Script Extender Debug Console
*****
*                                     *
*          BG3 Script Extender Debug Console          *
*                                     *
*****

BG3Ext v16 built on Apr 30 2024 22:11:17
Game version v4.50.22.896 OK
ScriptExtender::Initialize: Starting
Library startup took 980 ms
We're already past App load, triggering CoreLibInit
LibraryManager::PostStartupFindLibraries() took 108 ms
u Registered custom server network protocolload
Entering server Lua console.work protocol
S >> S >> extender signature to ClientConnect: LUA VM reset.
S >> ng ExtenderHello to client context Game
S >> xtender support notification from host (version 1)reset.
S >> xtender support notification from user 65536 (version 1)
S >> ScriptExtender::OnGameStateChanged(): Unloading session
S >> ScriptExtender::OnGameStateChanged(): Unloading session
S >> ScriptExtender::OnClientGameStateChanged(): Loading game session
S >> alizing client with target context Game
S >> ::ExtensionStateBase::LuaResetInternal(): LUA VM reset.
S >> ScriptExtender::OnGameStateChanged(): Loaded module
S >> alizing server with target context Game
S >> mFunctionManager::ClearDynamicEntries(): Clearing dynamic functions
S >> ::ExtensionStateBase::LuaResetInternal(): LUA VM reset.
S >> ScriptExtender::OnGameStateChanged(): Loading game session
S >> tExtender::RestartLogging: Starting Osiris Runtime debug logging.
S >> Path=/home/satan/Desktop/OsirisLogs/Osiris Runtime 2024-05-02 17-19-57.log
S >> tExtender::OnAfterOsirisLoad: 146182 nodes
S >> sionStateServer::StoryLoaded()
S >> mFunctionInjector mapping phase: 1289 -> 1289 functions
S >> sExtender::OnInitGame()
S >> isterDIVFunctions END
S >>
S >>
  
```

2.3. Basic inputs

Replicate the following steps to familiarize yourself with basic inputs into the SE console:

2.3.1. In the console enter the following command. This will print "Hello World" on the console:

```
_P("Hello World")
```

```
S >>
S >>
S >> _P("Hello World")
Hello World
S >> _
```

2.3.2. Add an item to your inventory - Here: 10 Bars of Soap:

```
TemplateAddTo("d32a68ff-3b6a-4d83-b0c4-0a2c44b93ea9",
GetHostCharacter(), 10)
```

```
S >> TemplateAddTo("d32a68ff-3b6a-4d83-b0c4-0a2c44b93ea9", GetHostCharacter(), 10)
S >> _
```



You will not see any print statement this time.
Instead check your characters inventory



2.3.3. Get the HostCharacter uuid (the character whose portrait is activated - here: Astarion):

`_P(Osi.GetHostCharacter())`

This will print: `c7c13742-bacd-460a-8f65-f864fe41f255` to your console

You will need this command quite often to easily access your character so it's useful to keep it in mind!



Now that you are familiar with entering commands in the console, it is time to create your workspace



Only want to execute a few commands on the console and not interested in creating a mod?

You can stop here if you are following another guide or jump to [6.Osirix](#) for a few examples

3. Creating your workspace



For the next steps, please close your game.



This guide uses this [example workspace](#)  you can download from google drive. Using it will make it easier to follow the tutorial.

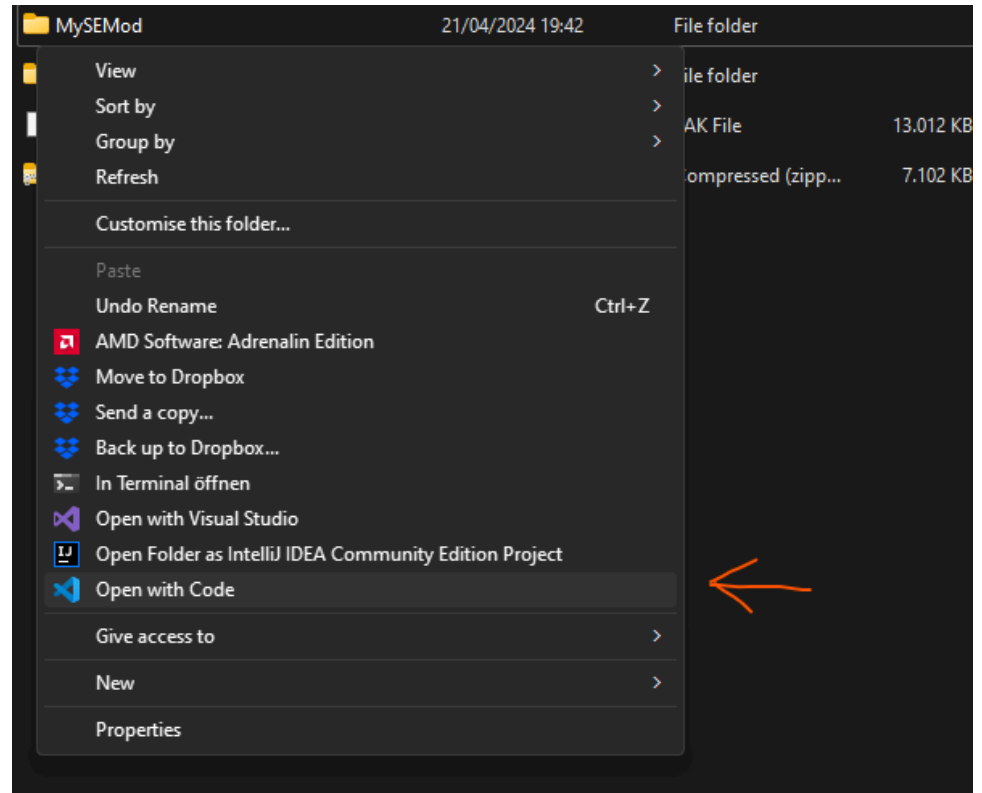
You will also need an editor. I recommend VSCode or the Open Source alternative VSCodium:

- VSCode: <https://code.visualstudio.com/> 

▸ VSCodium:<https://vscodium.com/> 

Download the example workspace linked above and unpack/unzip it.

3.1 Opening your workspace

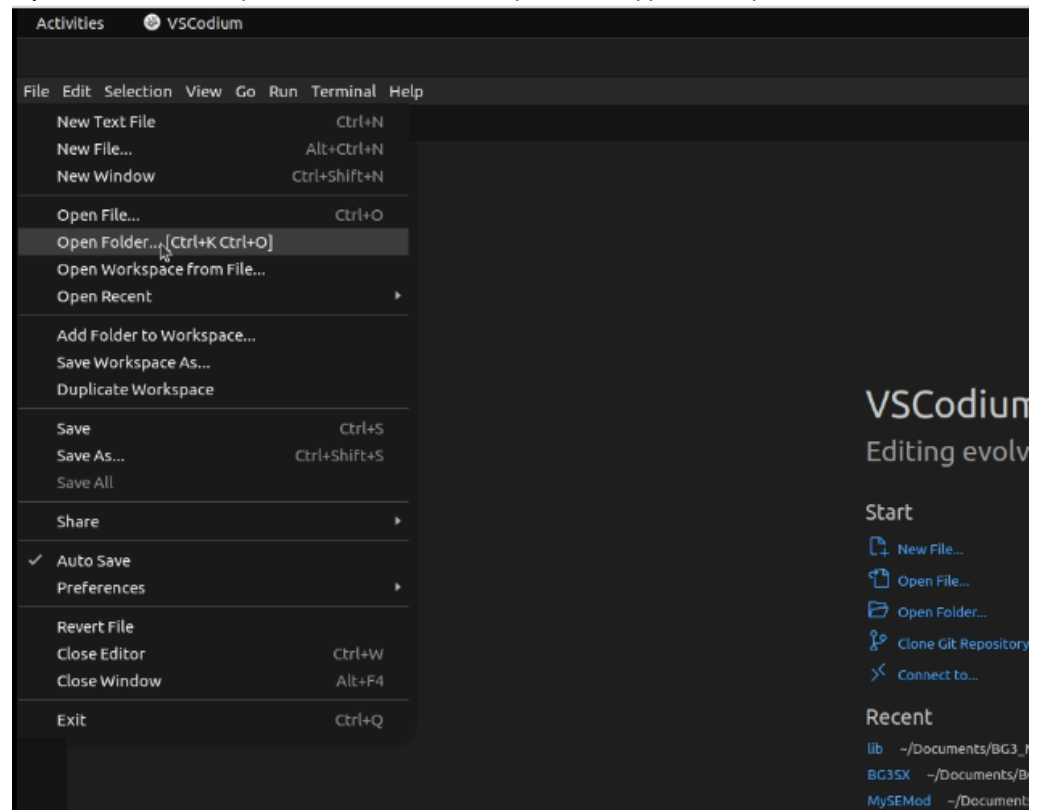


On Windows 11, right click the folder “MySEMod” after unzipping and click “Open with Code”.

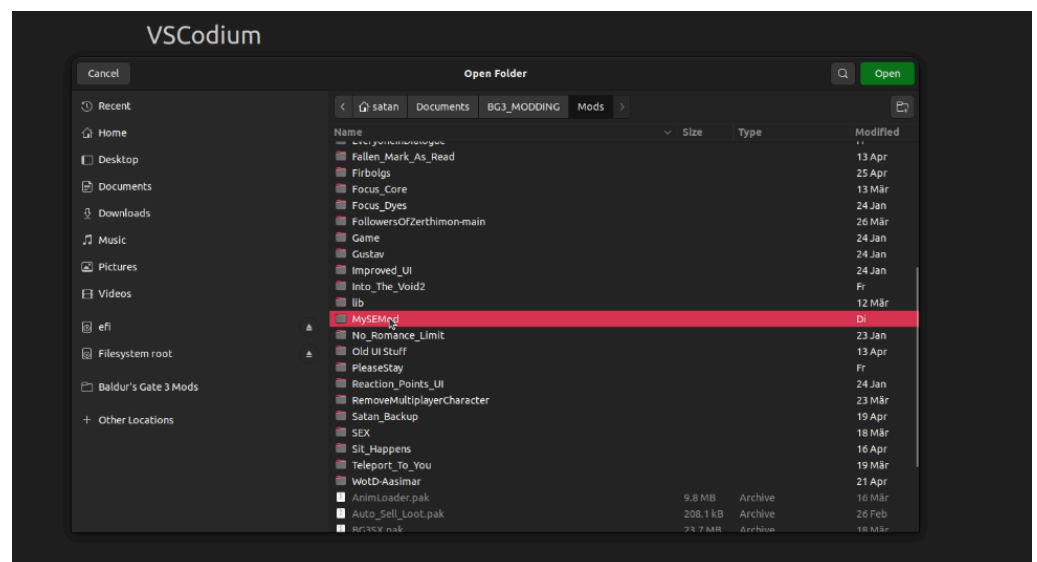


If you did not specify in the installation of VSCode that “Open with Code” should be added to the context menu in windows, you will not see this option. In this case follow the steps below instead.

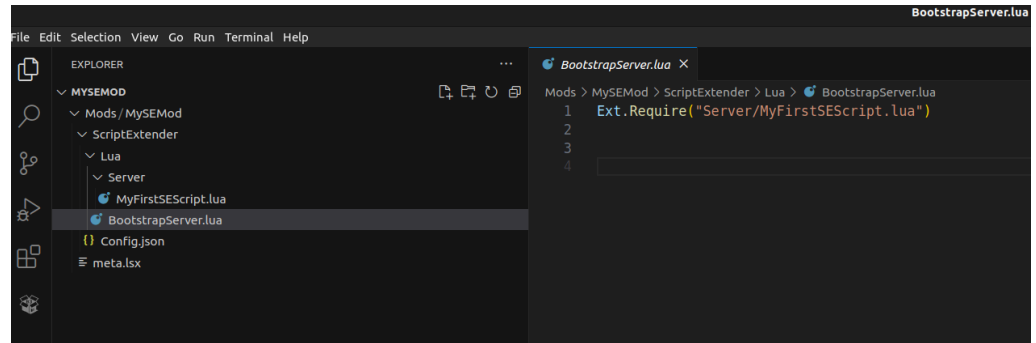
If you do not see that option, launch VSCode and open the unzipped workspace as a folder



Navigate to where you saved the folder and select it



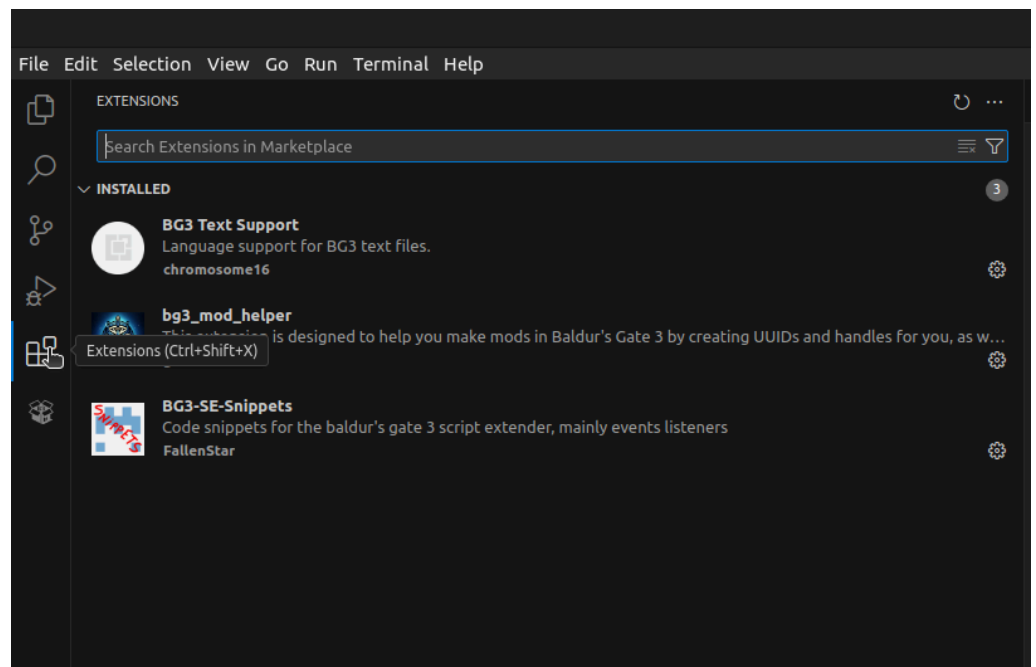
Now you should see the **MYSEMOD** folder in your editor. This is your workspace.



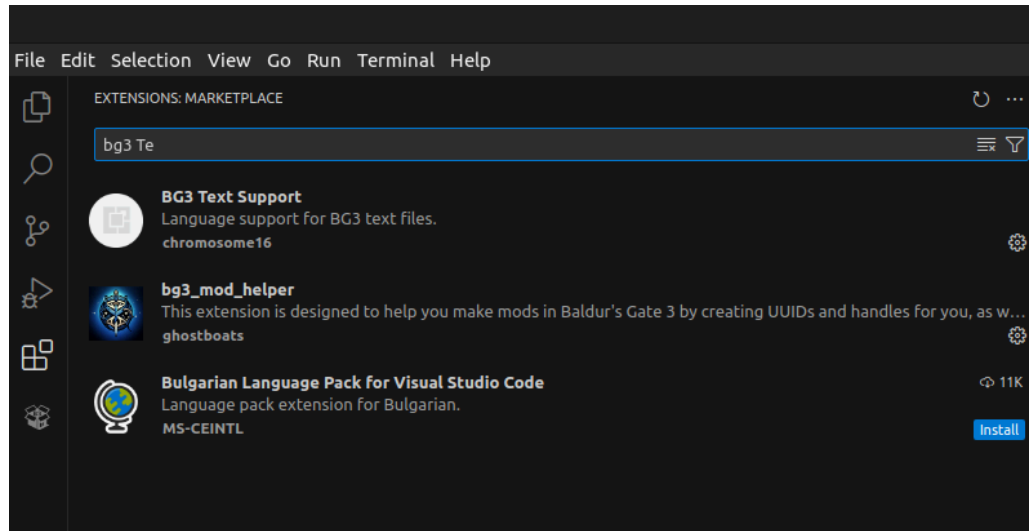
If you are planning on using this workspace as a template for a mod you want to publish you want to change the names and handle uuid in the `meta.lua` file. More information in [Dev Environment Set-up](#)

3.2 Installing Extensions

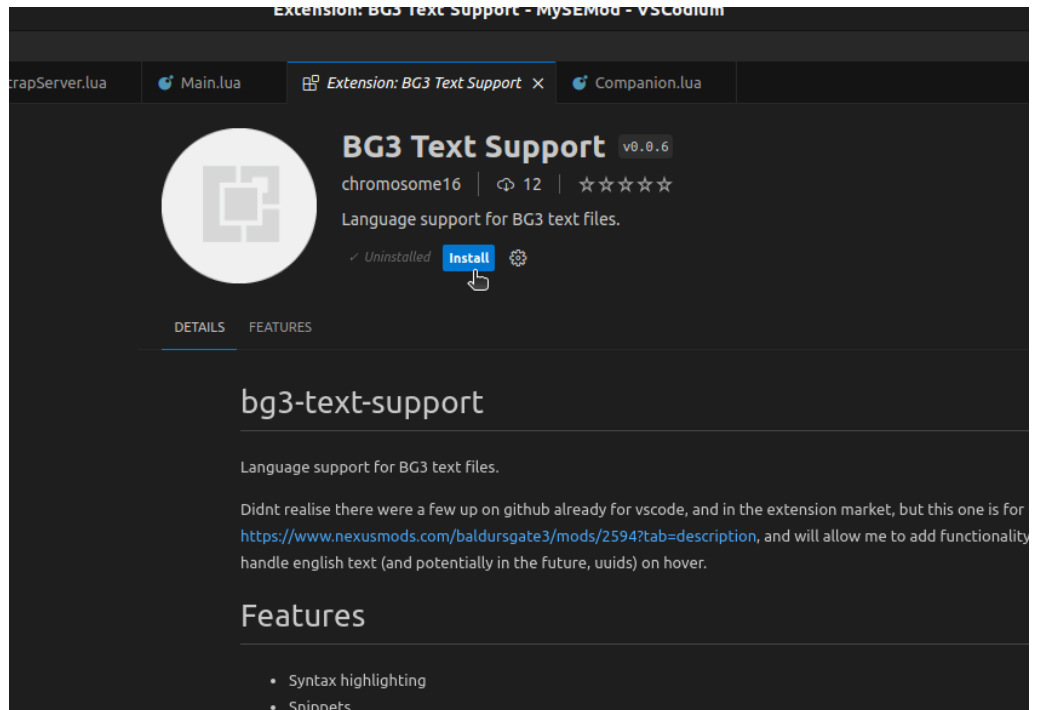
We will now install a few [Extensions](#) which will make creating mods a lot easier. For this click the [Extensions](#) button on the left. This will open a tab which allows you to see your installed [Extensions](#) and install new ones.



We will first install [BG3 Text Support](#) , [bg3_mod_helper](#) and [BG3-SE-Snippets](#) . These can be installed by searching for them in the search bar.



Click on the desired [Extension](#) and a new tab will open. Here click [install](#) .
Repeat this step for [bg3_mod_helper](#) and [BG3-SE-Snippets](#)

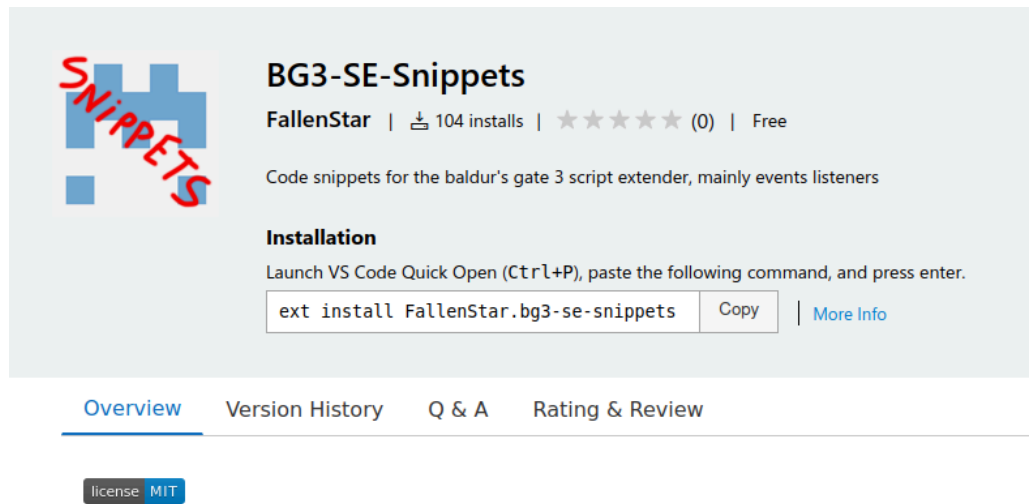


If you were not able to find [BG3-SE-Snippets](#) in the [Extensions](#) follow the manual install below:

Visit : <https://marketplace.visualstudio.com/items?itemName=FallenStar.bg3-se-snippets> [↗](#)

Follow the commands listed under **installation**. You can also see specific instructions in the next images.

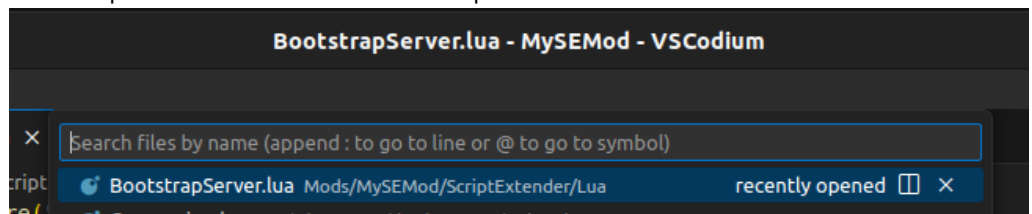
First, copy the text `ext install FallenStar.bg3-se-snippets`



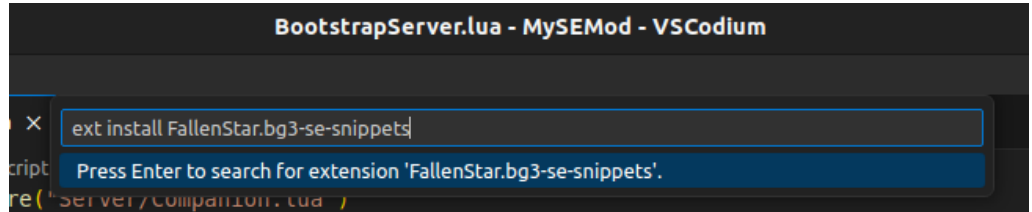
The screenshot shows the 'BG3-SE-Snippets' extension page. It features a logo with the word 'SNIPPETS' in red over a blue grid. The extension is by 'FallenStar', has 104 installs, and is free. The description states it provides code snippets for the Baldur's Gate 3 Script Extender, primarily for event listeners. Under the 'Installation' section, it instructs users to launch VS Code Quick Open (Ctrl+P), paste the command 'ext install FallenStar.bg3-se-snippets', and press enter. A 'Copy' button and a 'More Info' link are provided. Below the main content, there are tabs for 'Overview', 'Version History', 'Q & A', and 'Rating & Review'. A 'license MIT' badge is also visible.

Then open VSCode again and press `Ctrl+P`

This will open a small search window on the top



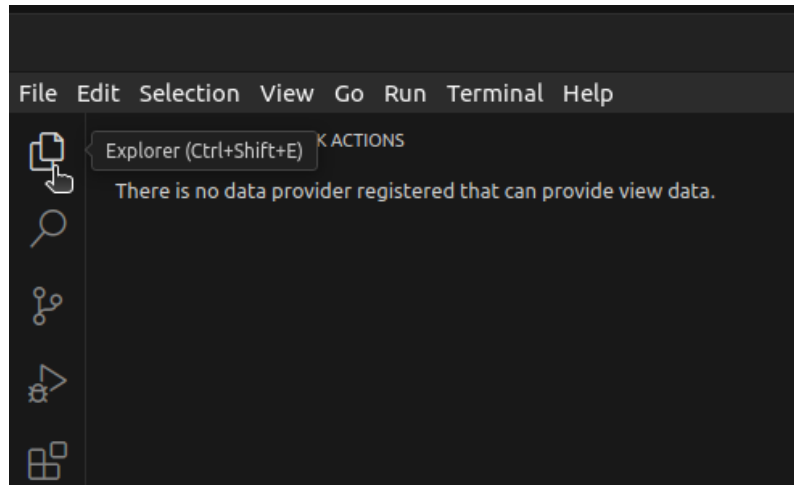
paste `ext install FallenStar.bg3-se-snippets` into this field and press enter.
This will automatically install this extension.



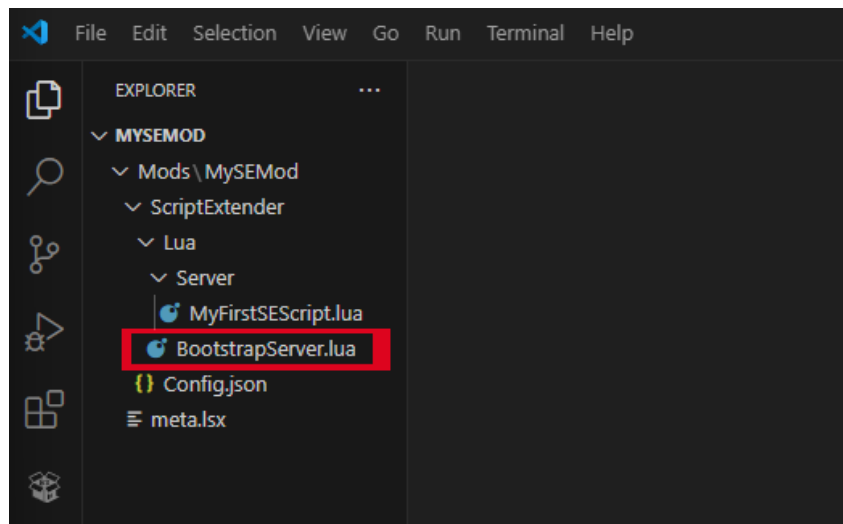
If you use VSCodium instead of VSCode you will have to manually install this extension

3.3 Adding your first line

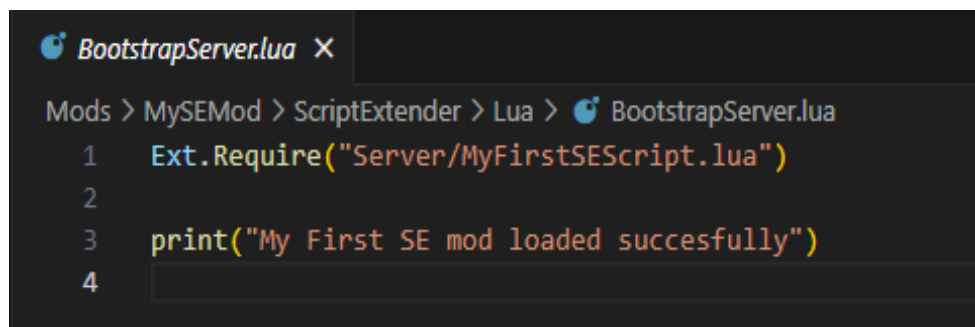
Now navigate back to your explorer by clicking [Explorer](#) .
You will find a few files here already.



Navigate to to `BootstrapServer.lua`



To test if your mod is active in the game, temporarily add a print statement that will execute when the game loads.



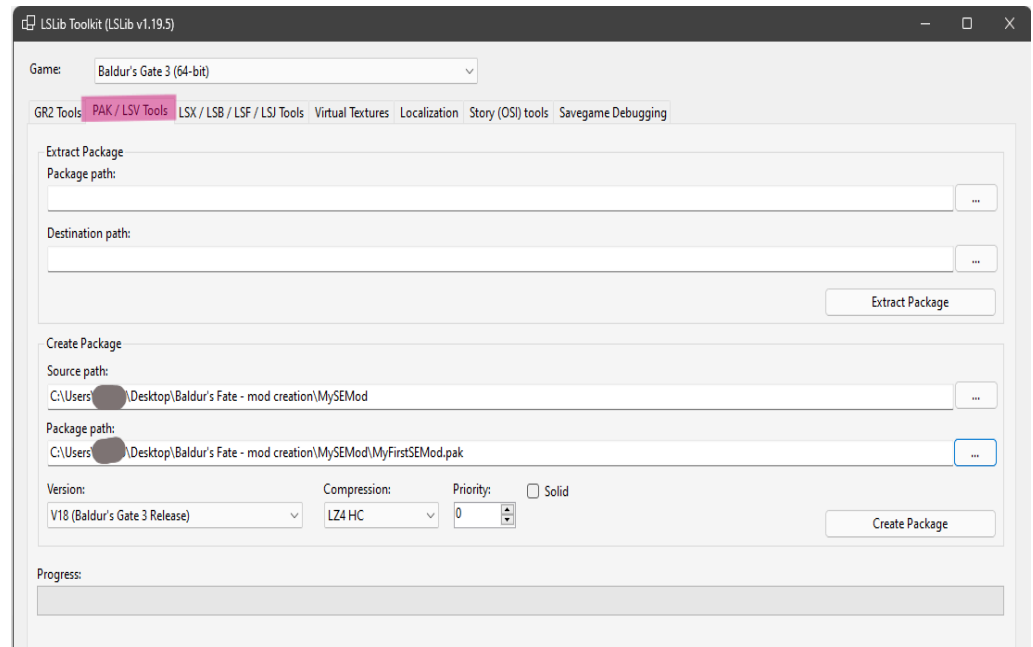
Adding this print statement causes Script Extender to execute it upon loading the game, letting us know if the mod is active.

3.4 Packing your mod

Afterwards, Pak the mod with a tool of your choice.

- ▶ LsLib: <https://github.com/Norbyte/lslib/releases> [↗](#)
- ▶ Modder's Multitool: <https://github.com/ShinyHobo/BG3-Modders-Multitool/releases> [↗](#)

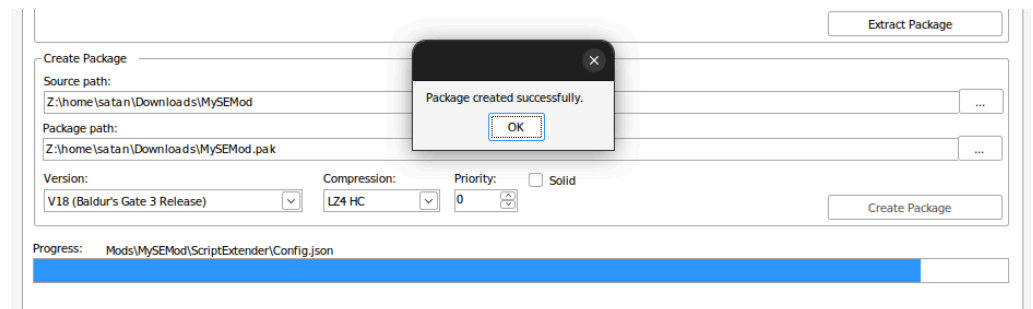
Here we will use lslib. The packing tool is called [ConverterApp.exe](#)
 Navigate to the correct tab [PAK/LSV Tools](#)



```

1 | MySEMod/ <- This is the folder you have to pak
2 | |
3 | |
4 | | └─ Mods/
5 | |     └─ MySEMod/
6 | |         └─ ScriptExtender/
7 | |             └─ Lua/
8 | |                 └─ BootstrapServer.lua
9 | |                     └─ Server/
10 | |                         └─ MyFirstSEScript.lua
11 | |
12 | | └─ config.json
13 | |
14 | |
15 | |
16 | |
17 | └─ meta.lsx
  
```

Pak the folder with a tool of your choice in the sections [Create Package](#)



Add the generated .pak file to your Mods folder and activate it and save the load order with your mod manager.

When you see your print statement on game startup, you have successfully loaded your mod. (You might have to create or load a save for the print to show up)

```

[common] [RECORD]: Registered user variable 'MyStringSend[...
Loading bootstrap script: Mods/MySEMod/ScriptExtender/Lua/BootstrapServer.lua
My First SE mod loaded succesfully
  
```

If you are not able to see this statement, you likely missed one of the steps.

4. Symlinking



If you want to symlink your mod, don't have a .pak version if it in your Mods.



As of patch 7 symlinked mods have to be activated and exported in BG3MM to be recognised.

Every time you make a change to your mod, you have to exit the game, repack your mod, and launch the game again to see the effect.

Symlinking is optional, although **highly recommended**, as it allows you conveniently avoid the aforementioned steps and hot reload your mod.



If this sounds too complicated, you can also move your workspace into the **Mods** folder in **Data**.

For this follow [4.2. Alternative to symlinking: Move your project to the Mods folder](#)

4.1. Setting up the Symlink (With a tool - Recommended for beginners)

Use [Link Shell Extension](#)  to simply create symlinks with the click of a button.

Check out their instructions [here](#) 

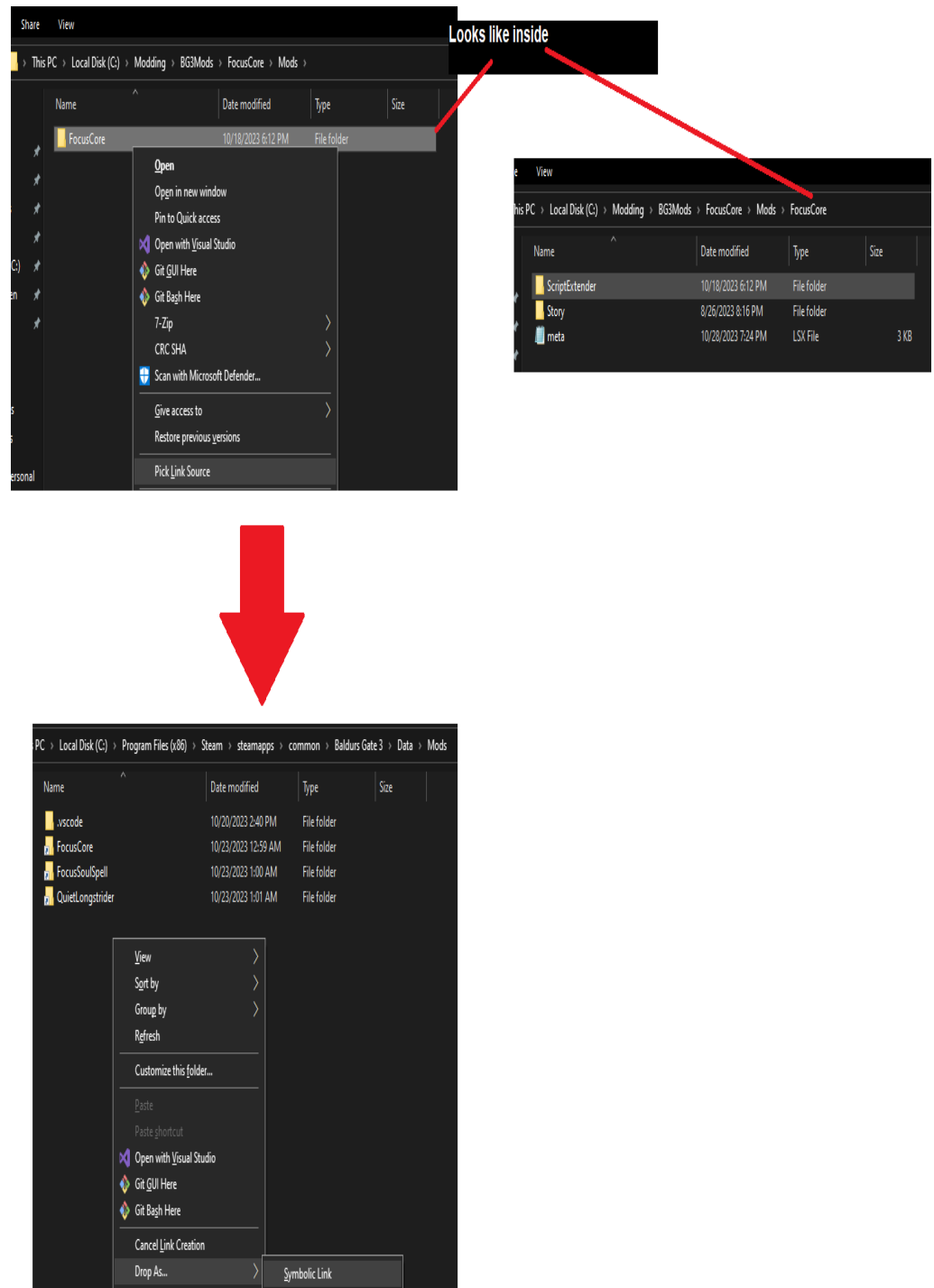


Image created by Focus and kindly provided by Muffin

4.2. Setting up the Symlink (Manual - For advanced windows users)

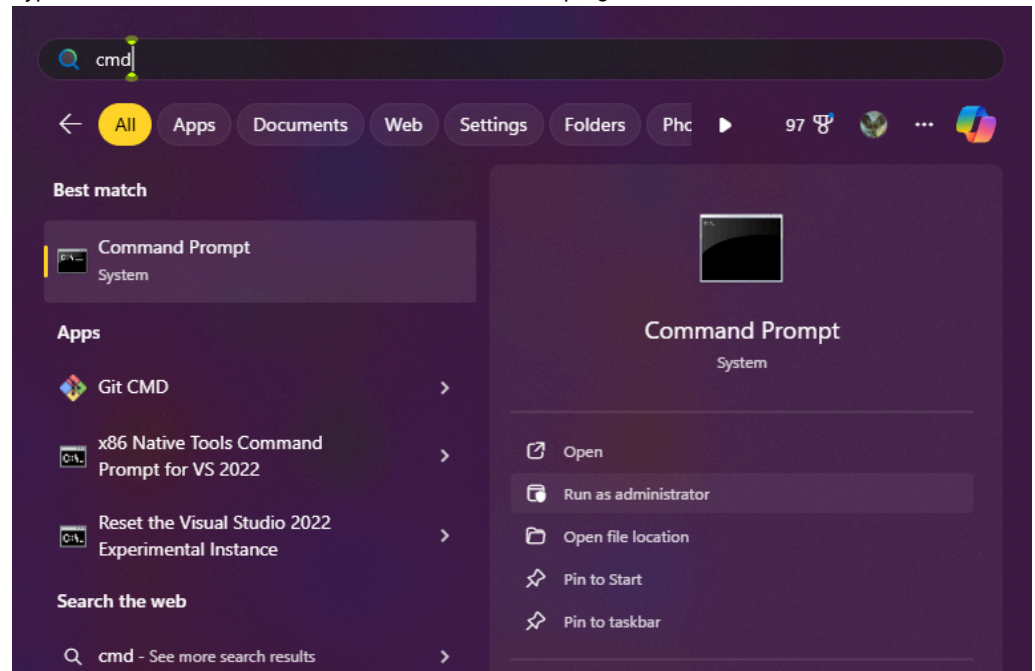
The following instructions are for Windows. On Linux, simply use `ln -s <workspace> <game files>`



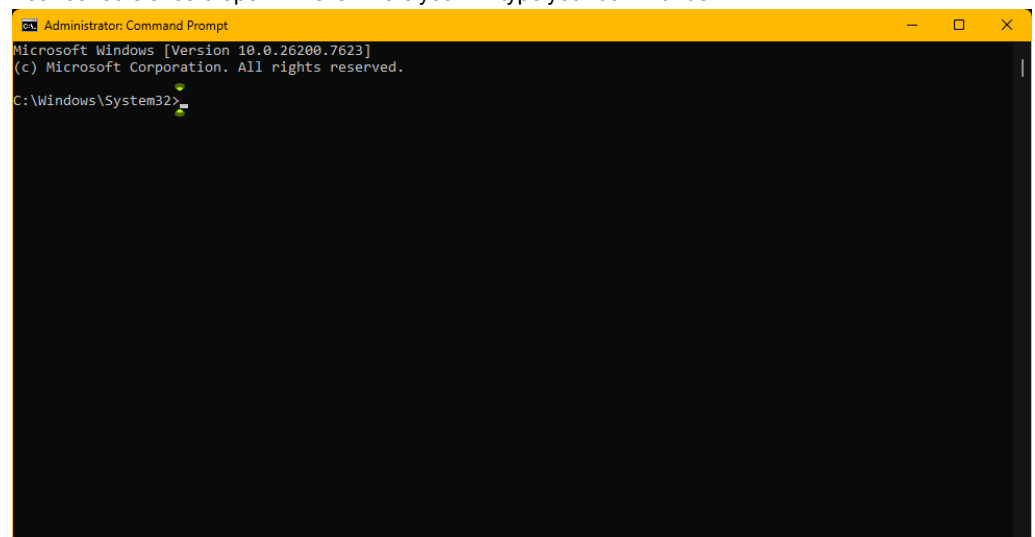
Due to wiki formatting, the commands had to be divided in separate lines.
When you input the commands, please make sure to write all in one line.

This command is executed in the windows terminal. Type `cmd` in the search bar and execute the program as **administrator**

Type `cmd` in the windows search bar and execute the program as **administrator**



Your console should open. This is where you will type your commands



Using `cmd` in **administrator** mode allows a lot of control over your system. Only execute commands here when you are certain about what they will do.

The commands for copying are found in the code block below. Please make sure to modify the commands, accordingly to your username, and direction of your workspace as well as your game. A text editor like **Notepad** is recommended to do this.



The folders **Mods**, **Public** and **Localization** do not exist in a Vanilla game. They will be created by the symlink.
If they already exist because you installed **loose files** mods, please delete them first, else the command will throw an error.

```

1 // Public folder
2 mklink /D "C:\Program Files (x86)\Steam\steamapps\common\Baldurs Gate
3 3\Data\Public"
4 "C:\Users\YourUsername\path\to\YourModName\Public"
5
6 // For code
7 mklink /D "C:\Program Files (x86)\Steam\steamapps\common\Baldurs Gate
8 3\Data\Mods"
9 "C:\Users\YourUserName\path\to\YourModName\Mods"
10
11 // Loca
    mklink /D "C:\Program Files (x86)\Steam\steamapps\common\Baldurs Gate
    3\Data\Localization\English\yourLoca.loc"
    "C:\Users\YourUserName\path\to\YourModName\English\YourLoca.loc"

```



Still stuck? Here are some other resources that explain this step differently
[Tutorial for Symlinking & Mass File Conversion for hot-testing your mods](#) [\[?\] \[?\]](#)



To delete the symlink simply delete the linked folder in
 \Steam\steamapps\common\Baldurs Gate 3\Data



An automated way of setting up your symlink might be provided in the future

4.3. Alternative to symlinking: Move your project to the Mods folder

You can also move your project to the **Mods** folder in **Data** instead if you were not able to symlink your folders.

C:\SteamLibrary\steamapps\common\Baldurs Gate
 3\Data\Mods\MySEMod

in the example below, where you can see the full path to the script it is:

D:\SteamLibrary\steamapps\common\Baldurs Gate
 3\Data\Mods\CBR_Hugs

DORIAN (D:) > SteamLibrary > steamapps > common > Baldurs Gate 3 > Data > Mods > CBR_Hugs > ScriptExtender > Lua > Server			
Name	Date modified	Size	
Hugs.lua	02.05.2024 22:19	17 KB	



Note that this symlink is for your scripts. If you want to include **.loc** or **.txt** / **.gr2** files
 They need to be placed in a **Localization** and **Public** folder respectively.
 For this follow the guide [How to install manual/loose file mods](#)

4.4. Verifying the symlink

To verify whether the symlink has been created in the correct space, click your newly created symlink in **Data**. This should lead you to your workspace folder.

If you see an error instead or a link to a subfolder, please repeat the steps in [4. Symlinking](#) and make sure your **paths** are correct.

4.5. Testing the symlink

To test whether your symlink has been successfully created, you can change the print line we have created earlier while having a save loaded.

```

BootstrapServer.lua X
Mods > MySEMod > ScriptExtender > Lua > BootstrapServer.lua
1 Ext.Require("Server/MyFirstSEScript.lua")
2
3
4 P("My First Script Extender Mod loaded succesfully")

```

Change it to something else so you can recognize the change

```

BootstrapServer.lua X
Mods > MySEMod > ScriptExtender > Lua > BootstrapServer.lua
1 Ext.Require("Server/MyFirstSEScript.lua")
2
3 print("I changed the text!")
4

```

Then, after activating your SE console again, type `reset` in the console and press enter

You can only utilize the reset statement if you either symlinked your folder or moved it into **Data**
Else you need to repack your Mod so see changes.

```

S >>
S >>
S >> reset_

```

This will reset the console and all loaded scripts, allowing you to see instant changes of your code.

Depending on your code, some changes will only apply when the Game session is loaded
In that case simply reload the game (F8)

```

S >> reset
[2]s[Ha[3JResetting Lua states.
CustomFunctionManager::ClearDynamicEntries(): Clearing dynamic functions
CustomFunctionManager::ClearDynamicEntries(): Clearing dynamic functions
bg3se::ExtensionStateBase::LuaResetInternal(): LUA VM reset.
Loading bootstrap script: Mods/VolitionCabinet/ScriptExtender/Lua/BootstrapServer.lua
Loading bootstrap script: Mods/AVItemShipmentFramework/ScriptExtender/Lua/BootstrapServer.lua
a[38;2;0;255;255;48;2;12;12m[AV Item Shipment Framework]: Aether & Volitio's Item Shipment Framework version 1.3.0 loaded
Loading bootstrap script: Mods/LunesAnimationFramework/ScriptExtender/Lua/BootstrapServer.lua
Loading bootstrap script: Mods/SweatySex/ScriptExtender/Lua/BootstrapServer.lua
Loading bootstrap script: Mods/TutorialChestSummoning/ScriptExtender/Lua/BootstrapServer.lua
Loading bootstrap script: Mods/MinimapMovement/ScriptExtender/Lua/BootstrapServer.lua
Loading bootstrap script: Mods/MySEMod/ScriptExtender/Lua/BootstrapServer.lua
I changed the text!
LoadingAnimations
ExtensionStateServer::StoryLoaded()
S >> bg3se::ExtensionStateBase::LuaResetInternal(): LUA VM reset.
Loading bootstrap script: Mods/VolitionCabinet/ScriptExtender/Lua/BootstrapClient.lua
Loading bootstrap script: Mods/AVItemShipmentFramework/ScriptExtender/Lua/BootstrapClient.lua
Loading bootstrap script: Mods/LunesAnimationFramework/ScriptExtender/Lua/BootstrapClient.lua
Loading bootstrap script: Mods/MinimapMovement/ScriptExtender/Lua/BootstrapClient.lua
Loading bootstrap script: Mods/BG3SX/ScriptExtender/Lua/BootstrapClient.lua

```



It has been reported that [Modders Multitool](#) might get stuck during the indexing step in case symlinked folders are present.
If this is an issue for you, simply remove the symlink and create it again after the indexing is done.

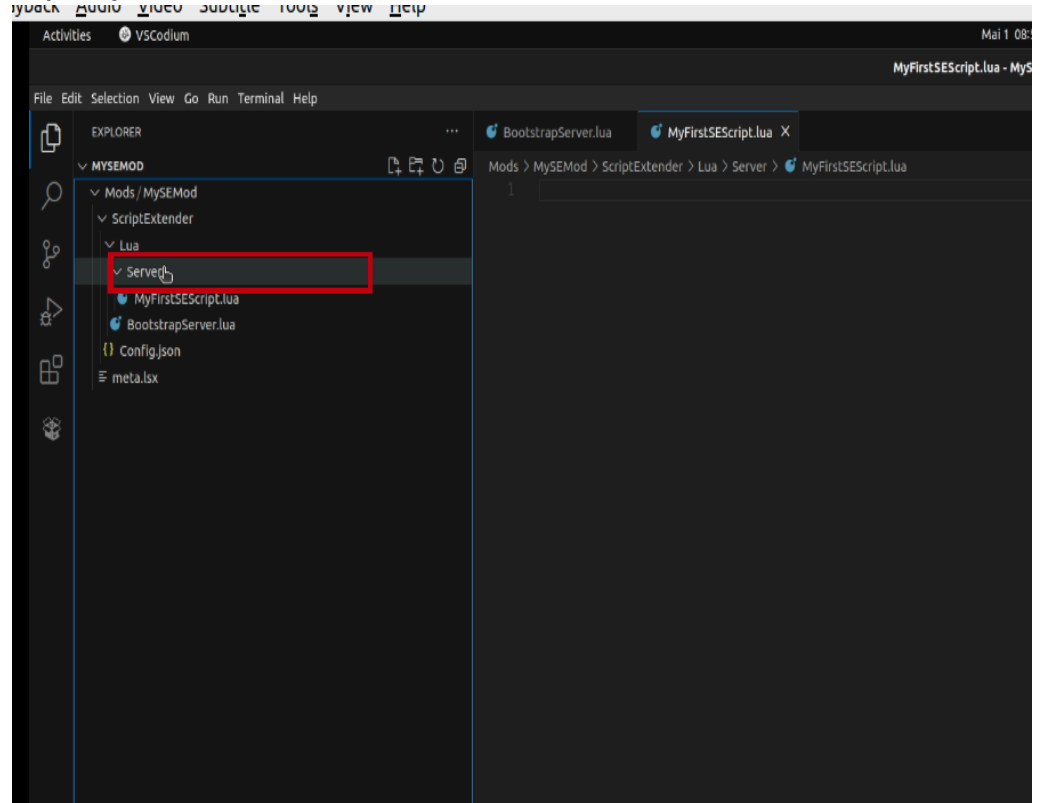
5. Creating multiple files

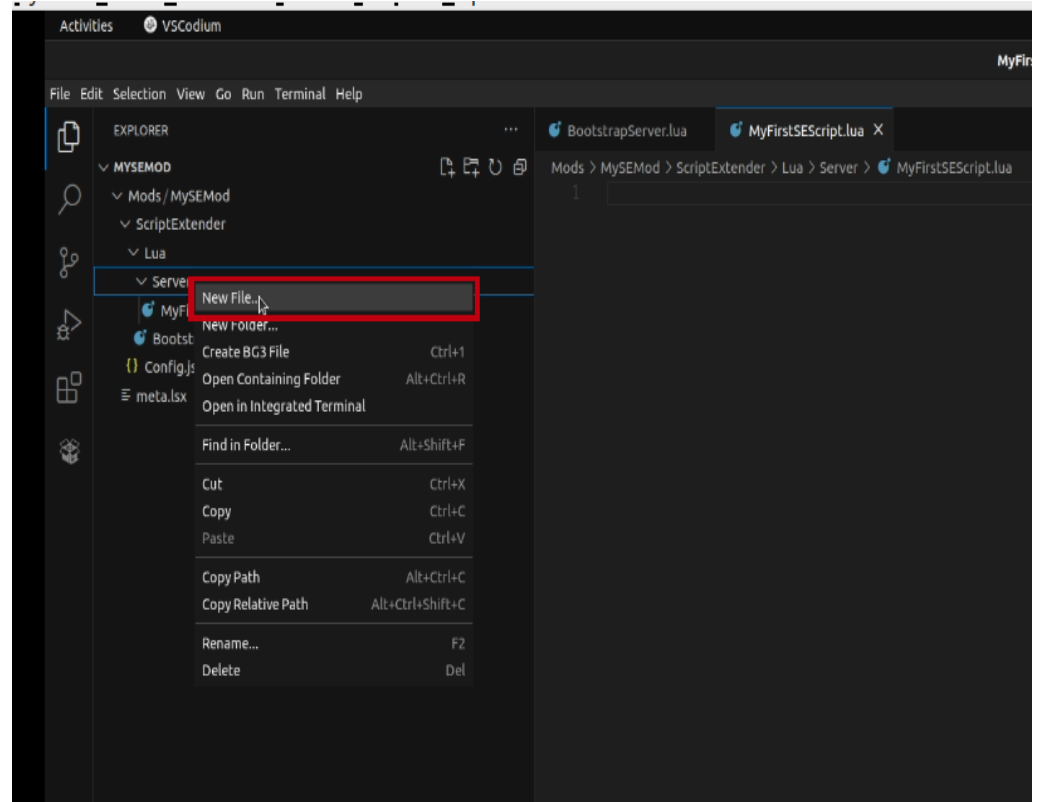
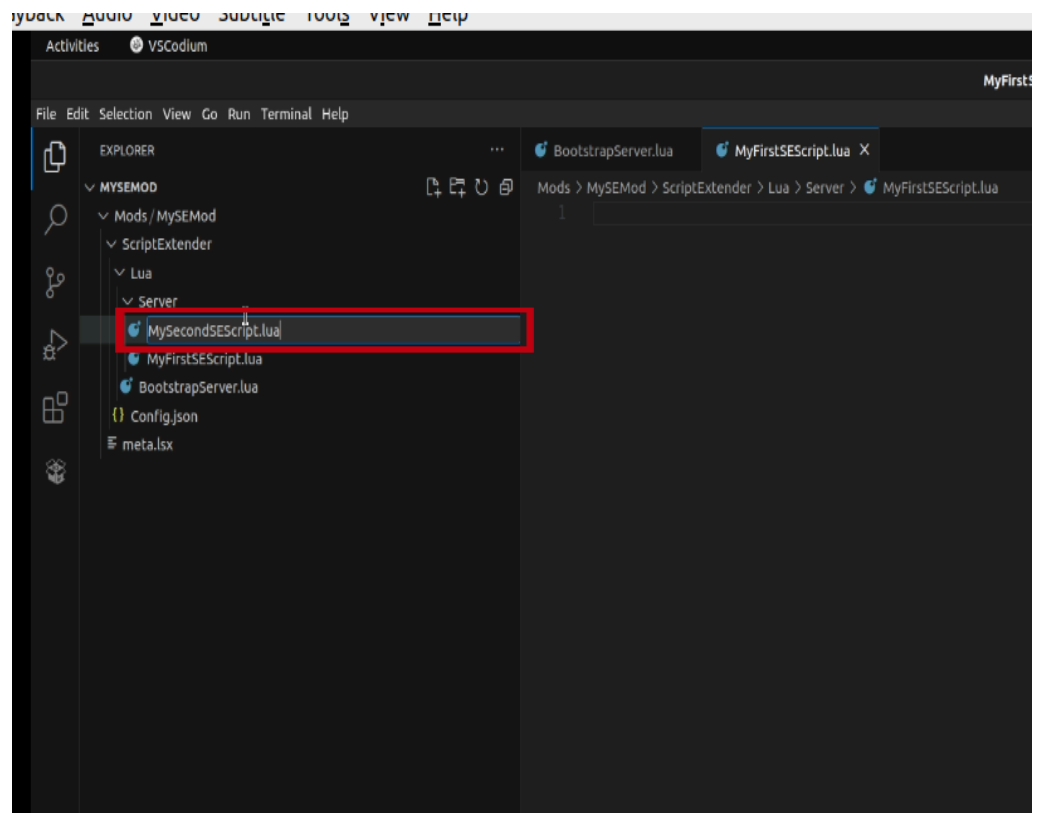
Whether you created a symlink or not, you can create multiple files in your workspace to help break down your project and for easier navigation.

Let us create a new file by right clicking on the folder “Server” and clicking “New File”.

We will name this file `MySecondSEScript.lua`.

Step 1: Right click on “Server”



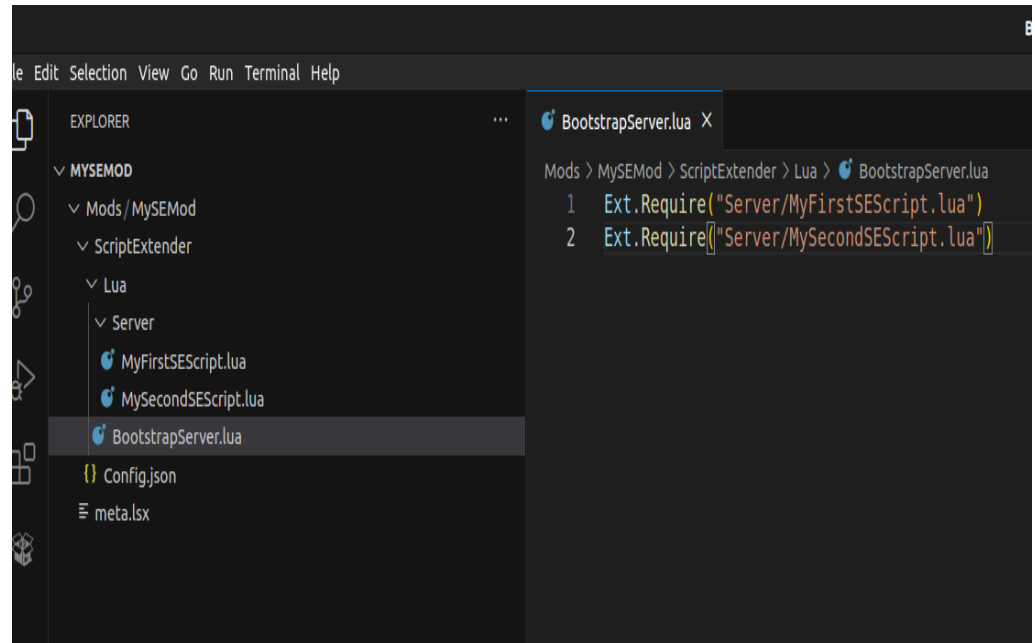
Step 2: Click "New File"**Step 3: Enter the name `MySecondSEScript.lua`**

All Lua scripts you want to create need to have the ending `.lua`

Step 4: Add the new file to `BootstrapServer.lua` by adding the line

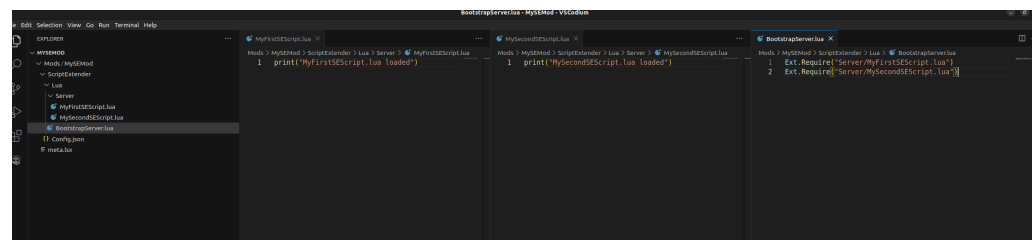
```
Ext.Require("Server/MySecondSEScript.lua")
```

Only by adding this line will SE know to load the new file. Else it will be ignored.



If you do not add your new file to the BootstrapServer.lua it and its contents will not be recognized

Step 5: Enter some print statements to make sure you can see the difference



Step 6: reset the console to see the difference

```
Loading bootstrap script: Mods/MySEMod/ScriptExtender/Lua/BootstrapServer.lua
MyFirstSEScript.lua loaded
MySecondSEScript.lua loaded
```



You can only utilize the reset statement if you either symlinked your folder or moved it into `Data`
Else you need to repack your Mod so see changes.

6. Osiris



The following section assumes that you have some basic programming and Lua knowledge

If you do no, please follow [The Basics of Lua](#)



Experienced programmers might still benefit from it as well since it introduces a few **Lua** quirks

Osiris is a programming language used by Larian Studios.^[1] Using Norbyte's Script Extender we can interact with Baldur's Gate 3 and manipulate it in a tremendous manner.^[2] Here we will have a look at a few simple examples. Specifically `Osi.lua`^[3] and `Osi.Events`^[4] provided by LaughingLeader^[5]. There are many more functions, but here we focus on the basics.

6.1. Functions

Here we will introduce a few functions from LaughingLeader's `Osi.lua` file. We will go over how we call the functions and how to use their output.

6.1.1. `Osi.GetHostCharacter()`

Here we will introduce one of the most important functions for testing.

`Osi.GetHostCharacter()` will return the **UUID** of the currently selected character

```
1 | ---@return CHARACTER character
2 | function Osi.GetHostCharacter() end
```

Here it is being called while Astarion is selected.

Since it returns a **UUID** we want to dump the return value with `_D` so we can see the output.

```
1 | _D(Osi.GetHostCharacter())
```

"c7c13742-bacd-460a-8f65-f864fe41f255"

6.1.2. `Osi.AddGold()`

Our next example will be adding gold to a character

```
1 | ---@param inventoryHolder GUIDSTRING
2 | ---@param amount integer
3 | function Osi.AddGold(inventoryHolder, amount) end
```

Here we can use `Osi.GetHostCharacter` for the `inventoryHolder` since we know that it returns a **UUID**.

For the second argument we need an **integer**, that is a non-decimal number.

```
1 | Osi.AddGold(Osi.GetHostCharacter(), 999)
```





6.1.3. Osi.GetApprovalRating()

We can not only execute functions that lead to changes in the game, we can also use them to get information.

```
1 | ---@param ratingOwner CHARACTER
2 | ---@param ratedCharacter CHARACTER
3 | ---@return integer rating
4 | function Osi.GetApprovalRating(ratingOwner, ratedCharacter) end
```

Here we will select out Tav so they will be the `host character` for the `ratedCharacter` .
For the `ratingOwner` we will choose Astarion. His UUID is
`"S_Player_Astarion_c7c13742-bacd-460a-8f65-f864fe41f255"`

Since here we `Get` a value we want to dump it again to see the output.

```
1 | _D(Osi.GetApprovalRating("S_Player_Astarion_c7c13742-bacd-460a-8f65-f864fe41f255", Osi.GetHostCharacter()))
```

1 (standard approval)

6.1.4. Osi.UseSpell()

We can also do more proactive actions with these functions.
Here we will cast a spell.

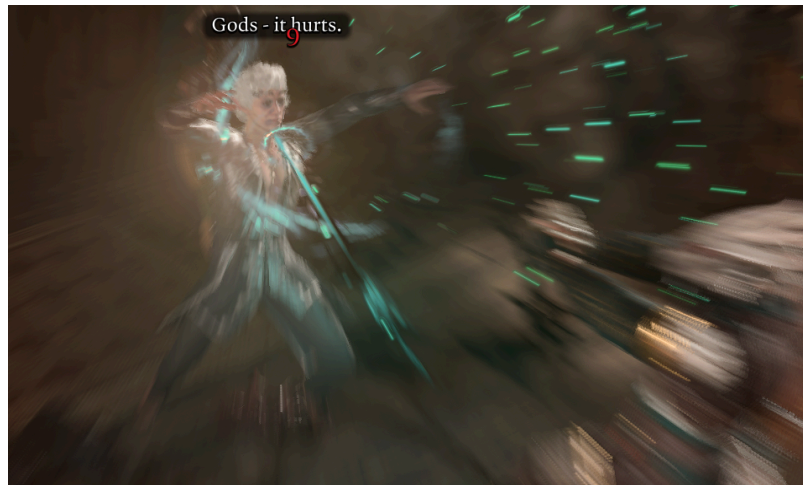
```
1 | ---@overload fun(caster:GUIDSTRING, spellID:string,
2 | target:GUIDSTRING)
3 | ---@param caster GUIDSTRING
4 | ---@param spellID string
5 | ---@param target GUIDSTRING
6 | ---@param target2 GUIDSTRING
   | function Osi.UseSpell(caster, spellID, target, target2) end
```

This function can be `overloaded` . This means that we can execute it with another set of parameters than listed. We can either use `caster, spellID, target, target2` or `caster, spellID, target` .

We are only interested in hitting one target right now so we will use the second option.

We will use our `Tav` to hit `Astarion` with `Eldritch Blast`

```
1 | Osi.UseSpell(Osi.GetHostCharacter(), "Projectile_EldritchBlast" ,
   | "S_Player_Astarion_c7c13742-bacd-460a-8f65-f864fe41f255")
```



Want to try out more spells? Use <https://bg3.norbyte.dev/search?q=type%3Aspell> to look up their names

6.2. Events and Listeners



Authors's note: This guide is still a work in progress. Please see [9. Useful Resources](#) for more information



Work in Progress. For now refer to Osi Events:
<https://github.com/LaughingLeader/BG3ModdingTools/blob/master/generated/Osi.Events.lua>



Work in Progress. For now refer to SE API Documentation:
<https://github.com/Norbyte/bg3se/blob/main/Docs/API.md>

For more interactivity we can use [Listeners](#) . If you followed [3.2 Installing Extensions](#) then you should have [BG3-SE-Snippets](#) installed. This will allow you to autocomplete listeners. Else you can build them yourself.

7. Script Extender Functions



Work in Progress. For now refer to SE IDE Helpers:
<https://github.com/Norbyte/bg3se/blob/main/BG3Extender/IdeHelpers/ExtIdeHelpers.lua>

```
// TODO
```

- Dumping
- Basically go over the API and heavily credit it

- List some useful Ext functions, like
_D(Ext.Entity.Get(GetHostCharacter()).CharacterCreationAppearance) - maybe even in a separate wiki page
- DB_Avatars, _D(Osi.DB_Players:Get(nil))
- Ext Listeners
- link to imgui page here

// TODO: Explain : and . (Calling the function with itself as the first parameter) -> Maybe something for 8. Using Script Extender Functions instead? Might fit in the metables section

<https://discord.com/channels/98922182746329088/1228009824017453147>  (ask for permission to post - original author is not clear)

8. Advanced Information

- IMGUI / Devel stuff?



Done reading?

[Create your first SE Mod with me](#) [TBA]

9. Useful Resources

Norbyte

Norbytes search engine: <https://bg3.norbyte.dev/> 

Script Extender: <https://github.com/Norbyte/bg3se/tree/main> 

SE API Documentation: <https://github.com/Norbyte/bg3se/blob/main/Docs/API.md> 

SE IDE Helpers:

<https://github.com/Norbyte/bg3se/blob/main/BG3Extender/IdeHelpers/ExtIdeHelpers.lua> 

LsLib: <https://github.com/Norbyte/lslib> 

FallenStar

FallenStar's VSCode extension: <https://marketplace.visualstudio.com/items?itemName=FallenStar.bg3-se-snippets> 

LaughingLeader

BG3MM: <https://github.com/LaughingLeader/BG3ModManager> 

Osi functions: <https://github.com/LaughingLeader/BG3ModdingTools/blob/master/generated/Osi.lua> 

Osi Events:

<https://github.com/LaughingLeader/BG3ModdingTools/blob/master/generated/Osi.Events.lua> 

Larian Studios

Osiris: <https://docs.larian.game/Osiris> 

DBTR Discord

[Down by the river](#) 

Me

Utils file:

Credits:

[Alithea Ancunin](#)  for proofreading

Cerberry for screenshots, ideas and feedback

Chip Chocolate, Legendary Muffin for a multitude of resources and his knowledge

ImmortalRD, SchroedingerCat and Chip Chocolate, Legendary Muffin for the [tutorial for Symlinking & Mass File Conversion for hot-testing your mods](#) 

[Padme4000](#)  for reporting the conflict between MMT and symlinks

simosas for his millions of useful functions and good company.

Skiz for bravely enduring and suffering through this journey with me

BG3 Modding Community Discord for answering my questions

References

[1] Osiris: A scripting language by Larian https://docs.larian.game/Osiris_Overview 

[2] Norbyte's Script Extender: An API that allows Lua/Osiris scripting for Baldur's gate 3
<https://github.com/Norbyte/bg3se> 

[3] Osi.lua: Osiris calls provided by LaughingLeader

<https://github.com/LaughingLeader/BG3ModdingTools/blob/master/generated/Osi.lua> 

[4] Osi.Events: Osiris events provided by LaughingLeader

<https://github.com/LaughingLeader/BG3ModdingTools/blob/master/generated/Osi.Events.lua> 

[5] LaughingLeader: Github link <https://github.com/LaughingLeader> 

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